Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/04/2018
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	02/29/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G appended with that information per discussions with M. Sansoucie & L. Strutzenberg/MSFC (Ed., 6/21/19) NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/28/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0256 ; NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC). Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	ISS expedition 56 PI (SCE FeCrNi), MUSC Control Center DLR Köln DE, 23 August 2018. Conducted MSL-EML flight space testing with Georg Lohöfer of DLR to investigate resistivity measurement protocols compare to previous parabolic flight results to compare cup and cage sample containment systems. ISS expedition 55 co-I (FeCo, TiAlCrNb) MUSC Control Center DLR Köln DE, 11-20 April 2018. Conducted investigations on magnetic alloy in cup to solve previous cage sample problems. Ran nucleation experiments on Ulm TiAl alloy with Rainer Wunderlich to compare to ground-based synchrotron results run with Olga Shuleshova of TU-Dresden. Conducted ground-based electrostatic levitation (ESL) flight support testing at MSFC January, March and May 2018 on future space CoSi and FeCrNi alloys. Sponsored NASA summer fellow Colby Azersky, a Tufts BSME senior, working at the Marshall Space Flight Center (MSFC) ESL under Michael SanSoucie on his BS thesis experimentation. Attended Investigator Working Group IWG#17 13-14 June 2018 at DLR Bonn Germany. Attended 2018 20th Properties for Materials Science at High Temperature Symposium as speaker, session chairman, and organizer at National Institute of Standards and Technology (NIST) Boulder. Attended ASGSR (American Society for Gravitational and Space Research) 2018 Annual conference as symposium organizer and presenter. EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC).
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Xiao X, Hyers RW, Wunderlich RK, Fecht H-J, Matson DM. "Deformation induced frequency shifts of oscillating droplets during molten metal surface tension measurement." Applied Physics Letters. 2018 Jul;113(1):011903. https://doi.org/10.1063/1.5039336 , Jul-2018

Articles in Peer-reviewed Journals	Matson DM, Xiao X. "Identifying metastable interface potency limits during steel alloy transformations." Materials Letters. 2018 Feb;212:256-8. https://doi.org/10.1016/j.matlet.2017.10.098 , Feb-2018
Articles in Peer-reviewed Journals	Matson DM. "Retained free energy as a driving force for phase transformation during rapid solidification of steel alloys in microgravity." NPJ Microgravity. 2018 Nov 19;4:22. eCollection 2018. https://doi.org/10.1038/s41526-018-0056-x ; PubMed PMID: 30480088; PubMed Central PMCID: PMC6242990 [Originally reported in Sept 2018 as "Accepted to npj Nature - Microgravity (2018) as of September 2018."] , Nov-2018
Dissertations and Theses	Nawer J. "Measurement of Mass Evaporation of Ni-Based Single Crystal Super Alloy Using Electrostatic Levitation (ESL) & Electromagnetic Levitation (EML)." MS thesis, Mechanical Engineering Department, Tufts University, May 2018. , May-2018
Dissertations and Theses	Cleaver A. "Characterization of the Influence of Thermal Profile on Microstructural Development in a Eutectic Cobalt Silicon Alloy." MS thesis, Mechanical Engineering Department, Tufts University, May 2018. , May-2018
Dissertations and Theses	Pleskus M. "Predictions for the Evaporation of Elements within Inconel 718." BS honors thesis, Mechanical Engineering Department, Tufts University, May 2018. , May-2018
Dissertations and Theses	Brienza M. "Resistivity Evaluation of Levitated Superalloys in Microgravity." BS honors thesis, Mechanical Engineering Department, Tufts University, May 2018. , May-2018